



2025

H22

BOOKLET NO.

134051

Botany
Paper – I

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

P.T.O.

**SECTION – A**

- Q1.** Write on the following in about **150 words each**. **(5×10=50)**
- Q1.** (a) Define disease modelling. Explain the different types of models used to understand and predict disease dynamics. **10**
- Q1.** (b) What is disease forecasting ? Explain the models and systems used in disease forecasting. **10**
- Q1.** (c) Explain the evolutionary developments in algal structure. **10**
- Q1.** (d) Give an account of economic importance of Fungi. **10**
- Q1.** (e) Discuss how algae are useful in industries. **10**
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- Q2.** (a) Give a detailed account of utility of beneficial microbes in agriculture and industry. **20**
- Q2.** (b) Explain Stellar evolution in Pteridophytes. **15**
- Q2.** (c) Elaborate the salient features of cycadales. Add a note on economic importance of cycadales. **15**
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- Q3.** (a) With the help of suitable diagrams, explain sporophytic evolution in bryophytes. **20**
- Q3.** (b) What are the different strategies adopted by parasites for nutrient acquisition ? **15**
- Q3.** (c) What is geological time scale ? Explain its major divisions. **15**
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- Q4.** (a) Explain the different ways of forming a petrified Fossil. **20**
- Q4.** (b) Explain the signal transduction pathways in plant defense. **15**
- Q4.** (c) What are the different forms of lichens ? Add a note on economic importance of lichens. **15**
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**SECTION – B**

Q5. Write on the following in about **150 words each**. **(5×10=50)**

Q5. (a) Write a note on International Code of Botanical Nomenclature (ICBN). **10**

Q5. (b) Enlist the centres of origin proposed by Vavilov (1926). Explain Indian centre in detail. **10**

Q5. (c) Give an account of different steps involved in a micropropagation technique. **10**

Q5. (d) What is numerical taxonomy ? Explain in detail the concept and steps involved in numerical taxonomy. **10**

Q5. (e) Enlist the taxonomically important chemical compounds in plants. With the help of suitable example explain the use of any one compound in chemotaxonomy. **10**

Q6. (a) Explain the advanced features of Family Orchidaceae. Add a note on its economic importance. **20**

Q6. (b) What are the major spices used in India ? Explain the chemical constituents and uses of any one spice. **15**

Q6. (c) Discuss the various methods employed for embryo rescue. Add a note on its applications. **15**

Q7. (a) Explain the concept of energy plantations. With the help of suitable example, explain the things needs to be considered for raising energy plantations. **20**

Q7. (b) Differentiate between the anatomical features of C_3 and C_4 plants. Draw suitable diagrams wherever necessary. **15**

Q7. (c) Write a note on protoplast culture. Add a note on applications of protoplast culture. **15**

Q8. (a) What are somaclonal variations ? How are they generated ? Give their advantages and disadvantages. **20**

Q8. (b) Explain the patterns of embryo development in plants. **15**

Q8. (c) Discuss the concept of Botanical Gardens. Explain any one Botanical Garden of India. **15**

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